

Mine in a box™

Managing Operations

Part A · Superintendent Handbook

Systems that make good decisions inevitable.

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1. Welcome to the program

Managing Operations runs a mine's production from the shift up. Your supervisors run the shifts. You run the department's days, weeks and month, and you coach the supervisors to run their shifts to plan.

Section 1 · Why you go first

Section 2 · How the program runs

Why you go first

You are trained at the same time as your supervisors, one step ahead of them. You go first, so that every workbook you sign and every card you coach from comes from someone who has worked through their own model.

How the program runs

Week	You	Your supervisors
Before week 1	The audit, with the Project Manager: the shift scored as it runs today	Not yet
1 and 2	Your Prime workbook: your day as it runs today. The Project Manager signs it. Then the Superintendents' Lab, half a day, where you get this handbook and your Apply tools	Not yet
3 to 5	Your first Apply: the Drive-around, the touch-base and the check-in every shift; each supervisor's workbook read and signed	Prime: the workbook on shift, one activity a shift
6	The Validation Lab: you coach from a card in the room	The Validation Lab: one day. They get their handbook and their Apply tools here, not before
6 to 8 weeks after the Validation Lab	Apply: each supervisor marks every card Yes or No, every shift; you coach from the cards, one No at a time	Apply: the shift run to the cards
After Apply ends	Verify: you observe the shift	Verify: one observed shift

"In this program we are teaching you how to manage. Management is the structuring of work and decisions so that performance remains predictable under pressure." – Dawid Falck

Where the learning happens (share of the program)



Continuous Mastery Loop™



2. What is work for you?

This chapter sets out your work: your job, where it stops and the supervisor's starts, how to manage and how you coach it, the decisions placed before the work, the assumptions the site was priced on, what your supervisors' shift looks like, and the decisions and data that pass through the day and the week.

Section 1 · Your job

Section 2 · Your job versus the supervisor's job

Section 3 · The definition of management and the six behaviours

Section 4 · Coaching the supervisors through the program

Section 5 · Decisions before the work

Section 6 · The assumptions the site was priced on

Section 7 · The supervisor's shift: the 44 lines

Section 8 · The decision and data map: day and week

Your job

You own the days, the weeks and the month. You place decisions before the work, so that your supervisors can run the shift without deciding the same things again every day.

The questions you ask, all week

- If we dig this block rather than that one, what happens to the two-week plan?
- Short of people, how do I order the work in this pit to hold our face positions?
- How is each of my supervisors travelling?
- Which decisions do my supervisors keep making on shift, and what rule or standard would place them before the work?

You are the supervisors' manager and the operators' manager once removed. You coach each supervisor at the Drive-around and the final drive-around. You fix what the shift cannot fix: the standard, the roster, the cab that stays broken. The supervisor fixes what can be fixed this shift. You fix what keeps coming back.

Where you are. The handover and the pre-start, watching. Your own drive-around of the pit, after the pre-start. The Drive-around with each supervisor at the end of their Block 1. The Daily Department Review and the Daily Site Review, which you chair. The touch-base at 11:30 (preferably at the lookout) and the check-in at 14:30. In Block 4, in the field with your supervisors until the last bucket, then the final drive-arounds at the end of the shift. Once a week, the Weekly Site Review; once a month, the Monthly Site Review.

Keep your supervisors in the field. A supervisor spends at least 90% of the shift in the field. No supervisor should have to leave the pit to get a fix from you.

Your job versus the supervisor's job

Your job is **not** the same as that of your supervisors. You have two very different jobs running in parallel. You are responsible for creating the system they work in. You set the targets and the golden rules (with their input) and focus on the weeks and months. You work with the other Superintendents (drill and blast, maintenance and so on) to make sure all the bigger systems of work link up smoothly.

You do not get to fiddle with their shift and their people. If you have an issue, you talk to the supervisor. They do not get to work on the plans for the weeks to follow. Their job is this shift, this hour, the minutes. Respect flows both ways, and it starts with clarity on what is and what is not work for each of you.

Levels of work	Supervisor · level II	Superintendent · level III
Time span	3 months to 1 year.	1 to 2 years.
Your focus on shift	This shift and the next.	The day across crews, the week and the month.
Accountability / output	The number delivered or explained; an honest Shift Report.	A system that works; supervisors who grow; the department's number.
Decides	Who does what, where. What to correct. Which contingency to trigger. The number committed at the Drive-around. Stopping unsafe work. Any worker may. What escalates, and when. Continue, correct, prepare or escalate at the end of each block.	The week's plan. Rosters and crews. Standards, TARPs and golden rules. Spend. Which live decisions become rules.
Does not decide	The plan and targets. Rosters. Hiring. Standards. Spend.	How the shift is run. Anything inside the supervisor's authority.
The test	A fix inside this shift, within a contingency, is theirs. Anything else goes up.	A decision re-made upstairs means a rule is missing.
Where the time goes	At least 90% in the field.	Checking, coaching and fixing what keeps coming back.
Decisions before the work	Finds the decisions made live, again and again.	Turns them into rules and checks them at Block 1.
Sign-offs	Fills in the cards, self-evaluating each line Yes or No, and the Shift Report.	Coaches the supervisor on them, every shift.
Coaching	Is coached; coaches one thing per operator per shift.	Coaches each supervisor; one point, checked next time.
Assumptions	Says when the ground does not match the plan.	Checks the site's assumptions against actuals each week.

If the plan is overridden. Your supervisors are told that if anyone issues orders that do not match the agreed plan or the four-block process, they may say, "I'm sorry, this is not how we work around here", and, if they are forced to comply, escalate it to you. Back them. If it comes from above you, it goes to your manager.

The definition of management and the six behaviours

> **Management is defining work, allocating work, coordinating work and verifying work.**
>
> **Management is the structuring of work and decisions so that performance remains predictable under pressure.**

#	Behaviour	Do I...?	Core function
1	Plan	Do I know the target, the risks and the controls before the shift starts?	Define
2	Assign	Does every person leave pre-start knowing their task, machine, area and target?	Allocate
3	Watch	Do I see the work with my own eyes, in the field, every block?	Coordinate
4	Respond	Do I tell people straight away what was good and what must change, and help them do it better?	Coordinate
5	Fix	Do I find and remove the cause of a problem, not just the symptom?	Coordinate
6	Report	Do I report my shift in PAVA (Plan, Actual, Variance/Impact, Actions taken) and set up the next shift for success?	Verify

What the research found. The best managers spend more time watching the work, so the feedback they give matches what people really did (Komaki; see "The evidence behind the model", page 55). That is why your supervisors are in the field 90% of their shift, and why you are in the pit with them in Blocks 1 and 4: you cannot respond well to work you have not seen.

You coach and check the six. Each roster, every supervisor rates themselves Yes or No on the six behaviours, two ways: what they do on shift, and what they instil in their crew so it happens when they are not there. They bring the card to you (their Apply Tool 8). Together you agree one behaviour to work on, and what you will look for in the field. Both of you sign. The six are yours too: mark yourself on them the same way, across your day and your week.

Time in the seat. Check that each supervisor spends two hours a roster doing an operator's job, in the truck or the digger seat, and ask what they noticed.

When you do this, use: Apply Tool 8, the six behaviours card.

Managing safety: running a high-reliability shift

A high-reliability organisation (HRO) does dangerous work every day and still has very few serious failures: aircraft carriers, air traffic control, nuclear power plants. They do not get there by luck or by more rules. They get there by how their people think and act every day, on every shift.

This does not replace your site's safety program. It strengthens it.

Your supervisors run their shift to these five questions:

Principle	The question they ask	Where in their shift
Preoccupation with failure	"What did we get away with yesterday?" Write down the answer and follow it up.	Step 2, pre-start
Sensitivity to operations	"What is really happening at the work right now?" Engage the operator at every work area; each roster, two hours doing an operator's job.	Step 3, Blocks 1 to 4 in the field
Reluctance to simplify	"Why?" five times after anything we got away with. Stop at the system cause, not the person.	Step 3, Block 3 (or straight after an event)
Deference to expertise	"Who is closest to this, and what do they say?" The person who knows decides. A stop-work is always backed.	Any block, any decision on conditions
Commitment to resilience	"What nearly went wrong today, and has it been handed over?" Keep it open until fixed.	Step 4 Shift Report and Step 5 handover, in Block 4

You check these five with each supervisor once a roster, Yes or No, with evidence, on Drive-around card 4, the five HRO questions. Your own part is the pattern: read the safety signals across crews at the Daily Site Review, and bring what your department keeps getting away with to the Weekly Site Review.

Coaching the supervisors through the program

A manager is a performance coach. You coach your supervisors on their own shift, from the cards, through the four stages of the program.

Stage	What the supervisor does	What you do
Prime (weeks 3 to 5)	Fills in the Prime workbook on shift: the shift as it runs today	Run the Drive-around, the touch-base and the check-in on your own model. Read each workbook and sign that it is done. You do not mark it. No workbook, no Lab.
Validation Lab (week 6)	One day in the room, from their own pages	Coach from a card in the room: the supervisor gives the first Yes or No, and you coach it.
Apply (6 to 8 weeks after the Validation Lab)	Runs the shift to the cards, every shift, and marks every card Yes or No	Coach from the cards, one No at a time, with an owner and a date.
Verify (after Apply ends)	Observed on one shift, Yes or No against the cards	Observe the shift. Sign what you saw.

The seven-step coaching cycle

1. **Permission.** Ask to coach; agree what you will watch.
2. **Set-up.** Look at what good looks like together, first.
3. **Observe.** Watch the work against the card.
4. **Reflect.** Ask open questions; help them find the answer.
5. **Commend.** Name what went well, specifically.
6. **Improve.** Agree what changes, and the action.
7. **Commit.** Get their commitment; set the next coaching time.

Coaching from the cards

Each supervisor fills in the cards and marks each line Yes or No, every shift, as they self-evaluate. You coach from them. A No is the next thing to coach, not a fault to report: check it the next shift. A run of No on the same line is a coaching conversation, and a sign to look at the system before the person. Ask the same questions, the same way, every time: within months the work is organised so the answer is ready.

When you do this, use: Apply Tool 4, the supervisor's cards, and Tool 9, the coaching record.

Decisions before the work

Decisions live in three places. Before the work (placed), during it (made live) or after it (reviewed). A decision placed before the work is made once, by the right person, and written down. A decision made live depends on whoever is standing there, under pressure. Most of the variation in a shift comes from where decisions are made, not from how hard people work.

The more decisions you can put before the work starts, the less we depend on variability and confusion during the work.

Where	What lives there
Before the work: placed	Rules and golden rules, TARPs, contingencies with a name on them, work allocations, priorities, the minimum crew
During the work: made live	What no rule covers: a digger down with no contingency named, an allocation argued on the radio
After the work: reviewed	The Shift Report and the final drive-around: which decisions were made live, what they cost, and which should become a rule

Levels of work. Every role has a time span: the longest task it is accountable for. Put a decision at the wrong level and it either waits for a meeting that comes too late, or is made again every week by people who cannot see the whole.

Role · level	Time span	Focus at work	What they work on
Operator · I	Up to 3 months	Today	The task in front of them
Supervisor · II	3 months to 1 year	This shift and the next	Today's priorities: the best of where we are
Superintendent · III	1 to 2 years	The day across crews, the week, the month	This month's and quarter's priorities: the best of where we can be
Project Manager · IV	2 to 5 years	The quarter and the year	The site's plan and its people
Operations Manager or General Manager · IV to V	2 to 10 years	One to three years and more	Lasting outcomes for the business

Turning repeat decisions into rules

At the final drive-around, your supervisor puts forward a decision they made live, written down in Block 4. You rule on it. If it becomes a rule, a TARP, a contingency or a standing allocation, it is written down, checked at Block 1 and never decided again by whoever is on duty. A decision you take for a supervisor is practice taken from them.

Building the guard rails. Work actively with your supervisors to find the decisions they are making again and again that lead to different ways of running a shift, and turn them into golden rules, so everybody operates with the same guard rails. The better you get at this as a team, the less chance there is of a supervisor coming back on roster and finding a pit that looks and operates completely differently from the way they handed it over.

The test of a placed decision. Could a supervisor who has never met you run this shift the same way, because the decision is written down, not in your head?

When you do this, use: Apply Tool 10, the decision-rights worksheet. Ask three people, separately, who decides. Where the answers differ, that is the row worth the time.

The assumptions the site was priced on

Before the site was built, someone assumed a digging rate for each digger, a truck count, cycle times, payloads, a delay allowance, support hours and a labour ratio. Those became the rates, and the rates became the budget. If you do not know what was assumed, you cannot tell whether a variance is your department's performance or somebody's estimate. These assumptions become your PADAL numbers (page 45). Each week, check them against actuals ("What to do about it", below).

Find these numbers, then check them against reality

Assumption	The question to ask
Digging rate, by digger	What loads an hour, or bcm an hour, was priced for each digger, and are we at it?
Truck numbers and truck rate	How many trucks were priced on each digger, at what cycle time and payload?
Delay (SMU) allowance	What share of engine hours was allowed for delay on each type of machine?
Payload	What payload was assumed for each truck class, and does the spread of payloads sit inside the site's payload rule?
Ancillary factors	How many hours of support gear (dozers, graders, water carts) were allowed for every 1,000 bcm or tonnes moved?
Labour factor	How many people hours were priced for each machine engine hour?
Availability, by fleet	What availability was priced, and are we above or below it?
Haul distance and cycle time	What haul was planned, and has the dump or the dig moved it without the rate moving with it?

Worked example: two numbers that disagreed

A site found that the delay allowances it reported did not match the ones it was priced on: 10% on one digger class and 7% on a truck class. The estimate also had no time for refuelling at all. It assumed refuelling happened in the break between shifts, and nobody had checked that every unit could be refuelled in that window. Until the two sets of numbers are settled, every talk about performance is really an argument about arithmetic.

What to do about it

1. Get the estimate assumptions in writing. If nobody can produce them, that is your first finding.
2. Compare each one with actuals, over a period long enough to be real: not one week.
3. Where they differ, decide which one governs, and write that decision down.
4. Take the gap to the Monthly Site Review, through the Project Manager. It is a commercial issue, not a production one.

When you do this, use: Apply Tool 11, the assumptions the site was priced on.

Your supervisors know these numbers for their own circuit, and say at the Drive-around when the ground does not match. That is where you hear it first.

The supervisor's shift: the 44 lines

The supervisor's shift has five stepping stones: Step 1, the handover; Step 2, the pre-start; Step 3, time in field (the four blocks); Step 4, time in office (the Shift Report and other admin, inside Block 4); Step 5, the handover to the next crew, also inside Block 4. Then the next crew goes back into the field.

The shift is 44 lines. Steps 1 to 4 each have a card; Step 5 uses the handover card, so lines 1 to 12 are marked at both handovers. Card lines continue across the cards. The supervisor marks every line; you coach from the lines where you see the work. Each card is a stand-alone tool in the supervisor's Apply tools. Their shift reports in PAVA (Plan, Actual, Variance/Impact, Actions taken), in loads and buckets per hour.

Lines 1 to 22: the handover and the pre-start

Time/hours	Supervisor's step	Lines	Where you see them
05:00	Step 1. Handover · before Block 1	1 Started and finished on time, with all supervisors and Superintendents there; meeting behaviours kept. 2 The plan for this shift was ready before 05:00 and used in the handover. 3 Last shift taken over in PAVA from the report: dig position, rates, hazards, what was escalated, work not finished. 4 Pit plan and layout checked in the office against the mud map or pit map, area by area; understanding checked. 5 Targets set for the staff and gear really there this shift, not the budget or an average. 6 The focus or priority areas for the shift agreed. 7 The contingencies in force reviewed, with the TARPs and golden rules; each contingency has a name. 8 The next 24 hours checked across drill, blast, load and haul, ROM and maintenance: services, supplies, gear and order of work. 9 Each risk to the plan either planned around or escalated, with a name on it. 10 Told what this shift must set up for the next. Checked what the last shift left set up. Each item has a name. 11 What did not fit was challenged; changes agreed with the Superintendent before pre-start. 12 Key actions written on the plan as tasks for the crew, ready for pre-start.	In the office, at the handover
06:00	Step 2. Pre-start · before Block 1	13 Started and finished on time, with the whole crew there; meeting behaviours kept. 14 Run from the plan made at the handover, checked by the supervisor before the meeting. 15 Safety first: today's hazards, areas and safety focus, in more detail when something has changed or the crew is new. 16 Last shift's results given briefly, area by area. 17 Each area and its key work described; every operator given a machine, an area and a task. 18 The shift target said to the crew in loads and buckets, with first and last bucket times. 19 Operator scorecard up to date and on the wall where all can see it; at least once a roster, the top performers named and congratulated, on the roster's trend. 20 Spoken slowly and clearly, with authority and energy. 21 Understanding checked by name: operators can say first and last bucket, the targets, the contingencies, the priority areas and the risks to the numbers; a safe and productive shift expected. 22 Crew out of the room within 30 seconds of the end, straight to their gear.	Watching the pre-start

Lines 23 to 44: the four blocks, the Shift Report and the handover to the next crew

Time/hours	Supervisor's step	Lines	Where you see them
06:15	Step 3. Time in field · Block 1 · Start up	23 First bucket on time; all operators to their gear on time. 24 Go-lines operating efficiently; fuel plan in place and kept to; all work areas inspected and set up for safety. 25 The handover you got verified as correct. 26 The shift target confirmed as achievable with the crew actually on shift, the gear available and the pit conditions as expected.	At the Drive-around
09:30	Step 3. Time in field · Block 2 · Execution	27 Loads, buckets, payloads from the operators' counts, beside the system's; differences settled. 28 Top delay acted on; one safety interaction at the work. 29 Touch-base 11:30, in numbers: continue, correct, prepare or escalate.	At the 11:30 touch-base
11:45	Step 3. Time in field · Block 3 · Optimise	30 Stopwatch on hang, queue, load and cycle time; what was needed changed and measured. 31 One operator coached at the work. 32 Check-in 14:30: the gap, and the shift left.	At the 14:30 check-in
14:45	Step 3. Time in field · Block 4 · Set up and hand over	33 Last bucket protected. 34 Next shift's areas inspected and set up. 35 What we got away with, what went wrong and the decisions to make before the work, written down.	At the final drive-around
By about 16:00	Step 4. Time in office · in Block 4	36 Written on the Shift Report sheet through the blocks, from the operators' counts, not from memory. 37 Plan against actual, in PAVA, in loads and buckets per hour, block by block and at the end, with any statutory reports. 38 Jobs not finished this shift are listed. 39 It shows what was done, and when, each time the shift went off plan. 40 The gap in one line; each variance explained (its impact, the action taken and whether it worked) or accepted in writing. 41 What could have foreseen or avoided each variance, and whether it reaches the next shift, written down. 42 Risks to the next shift written down, with the jobs it must plan in. 43 Done by about 16:00, before the push to the last bucket. 44 Explained at the final drive-around and used at the end handover with the oncoming supervisor.	At the final drive-around
16:30 to 17:00	Step 5. Handover to the next crew · in Block 4	Lines 1 to 12 again, on the handover card: the final drive-around first, then the handover to the next crew. The things that need to be escalated are clearly defined and agreed.	At the final drive-around and the 17:00 handover

The decision and data map: day and week

Meeting	Decides	Receives (from, by when)	Hands on (to, by when)
Handover, 05:00, in the office, with all supervisors and Superintendents	The priorities for the day; the contingencies for the day, to mitigate the identified risks; the allocation of gear and people; what this shift must leave ready for the next	Last shift's report, in PAVA (off-going supervisor, by 05:00); the things that were escalated and still need to be resolved	The plan to the pre-start (supervisor, 06:00)
Pre-start, 06:00	Fit to start, area by area	The plan and target (supervisor)	The crew out of the room by 06:15, straight to their gear
Drive-around, 09:00	The number for the shift; the priority for a short crew	Block 1 seen on the ground (supervisor)	The number and what cannot be fixed (you, to the Daily Department Review by 10:00)
Daily Department Review, 10:00	A fix for each issue a supervisor cannot fix; the department's line	Numbers and issues (Drive-arounds, by 10:00)	Fixes (supervisors, by 11:30); the line (Daily Site Review, 10:30)
Daily Site Review, 10:30	Collisions between departments; the next 24 hours; where help goes	Each department's line (Superintendents, 10:30)	Fixes back the same shift (supervisors, by 11:30); repeat collisions (Weekly Site Review)
Touch-base 11:30 and check-in 14:30	Each area: continue, correct, prepare or escalate	The block in loads and buckets per hour (supervisor)	Escalations (you, the same shift)
Shift Report, by about 16:00	The verdict on the shift; each variance's action or acceptance	Operators' counts (every block)	The report (you, by about 16:00)
Final drive-around, 16:30 to 16:50	Each variance explained by the supervisor: its impact, the action and whether it worked, and whether it reaches the next shift; one coaching point; one rule put forward	The four block cards and the Shift Report (supervisor)	Signed actions page (you); the key actions for the next crew (handover, 17:00)
Handover to the next crew, 17:00	What the next crew is handed; the things that need to be escalated, clearly defined and agreed; the key actions needed from it	The report, what was left set up, and the key actions (off-going supervisor)	The next shift's plan (oncoming supervisor); the crew back into the field
Weekly Site Review, weeks 1 to 3	Next week's threats, each with a name; the site's priorities; what goes up	Each department page (Superintendents, before the meeting)	The site's week (head office IN meeting, by the night before); flags for the Monthly Site Review; threats and changes (Weekly Site Planning Meeting)
Weekly Site Planning Meeting, weekly	Next week's plan: sequence and volumes by day; maintenance and blast windows locked	Threats and changes (Weekly Site Review); the week in PAVA	The week ready for signature (Weekly Commitment Meeting)
Weekly Commitment Meeting, weekly	The week signed by the people who will deliver it	The week for signature (2 to 3 hours before)	The signed week, published within 2 hours (every shift plan and handover)



3. The five stepping stones of your day

There are five things a Superintendent must do well to run the department's day. We cover those next.

Section 1 · The five stepping stones at a glance

Step 1 · Your four blocks

Step 2 · The Drive-around cards

Step 3 · The Daily Department Review

Step 4 · The Daily Site Review

Step 5 · Taking part in the Weekly Site Review

The five stepping stones at a glance

Your day has five stepping stones: your four blocks, the Drive-around cards, the Daily Department Review, the Daily Site Review and the Weekly Site Review. Your day runs on the same clock as your supervisors' shift. Where the two meet, something passes between you: the number at the Drive-around, the fixes after the meetings, the numbers at the touch-base and the check-in, the cards and the variances at the final drive-around.

Step	The stone	Block	Where
Step 1	Your four blocks	Blocks 1 to 4	The pit and the office
Step 2	The Drive-around cards: the Drive-around, the touch-base and the check-in	Block 1 (from 09:00), then 11:30 and 14:30	The pit, with each supervisor
Step 3	The Daily Department Review	Block 2, 10:00	The office
Step 4	The Daily Site Review (you chair it)	Block 2, 10:30	The office
Step 5	Taking part in the Weekly Site Review	Weekly	The office

The cards are your cheat sheets. Each coaching card is a stand-alone tool. Take it from your Apply tools and take the card for each step with you: you do not need to carry this book on shift. You mark each card Yes or No yourself, as you self-evaluate, and bring them to your own coaching with the Project Manager.

Your Apply tools (issued with this handbook): Tool 1, your four block cards · Tool 2, Drive-around card 1 · Tool 3, Drive-around card 2 · Tool 4, the supervisor's cards · Tool 5, the Daily Department Review log · Tool 6, the Daily Site Review log · Tool 7, the weekly department page · Tool 8, the six behaviours card · Tool 9, the coaching record · Tool 10, the decision-rights worksheet · Tool 11, the assumptions the site was priced on.

Your day shift

Time/hours	Block	What happens	Steps
05:00 until the Drive-arounds are done, before 10:00	1 · Start up, observe, verify	The handover in the office and the pre-start, watched (after a night shift, the night supervisor explains the variances first); your own drive-around of the pit; then, in the office, people, the plan and the night shift's work; the combined Drive-arounds from 09:00, at the end of each supervisor's Block 1	1, 2
10:00 to 11:45	2 · The meetings	Daily Department Review at 10:00, 20 minutes; Daily Site Review at 10:30, 30 minutes; the touch-base from 11:30 (preferably at the lookout)	3, 4, 2
11:45 to 14:45	3 · People and plan	Recruiting, training, the two-week plan, each supervisor's coaching focus; the check-in from 14:30	1, 2
14:45 to 17:00	4 · Time in the field, final drive-arounds	In the pit; your supervisors' write-ups done by about 16:00; then in the field with each of them until the last bucket, at about 16:30; the final drive-arounds from 16:30 (with several supervisors, see "More than three supervisors, and the night shift" in Step 2); the 17:00 handover, with all supervisors and Superintendents	1
Weekly		The Weekly Site Review: you bring the department page; the Project Manager chairs	5

Times are indicative. Your site can change them. The order does not change.

The worked example. One example runs through every step, with illustrative numbers. North Pit department, day shift. The A crew supervisor runs one digger (EX1) and four trucks (T1 to T4) on the south bench: 12,000 bcm, 200 loads, about 22 loads an hour. A second supervisor runs the EX2 circuit.

Step 1 Your four blocks

Key question: Is the department set up to deliver today, and what keeps coming back that I can place before the work?

Instead of running your day from one call to the next, this program breaks it into four blocks, on the same clock as your supervisors' shift. Each block has one clear focus: check before the pit commits, get the fixes back before the shift is lost, work on what outlives this swing, and verify and coach on the ground. You run them; nobody runs them for you.

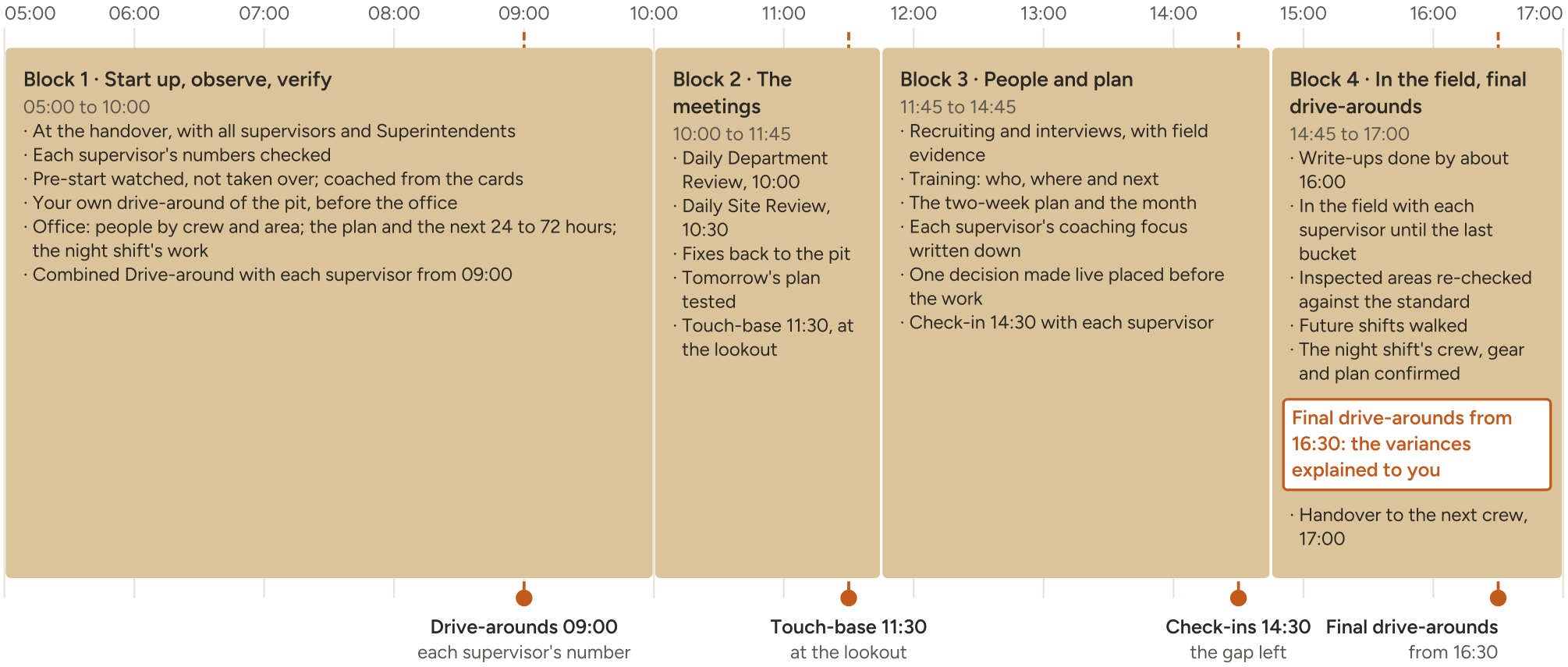
The blocks protect your supervisors' blocks as much as your own. Because everyone knows where you will be and when, the constant calls stop, and nobody has to leave the pit to find you.

Outcome: You are in the pit when your supervisors need you, in the office only after your own drive-around and in Blocks 2 and 3, and every supervisor ends the day coached on one thing, with one rule placed before the work and the next shift set up.

- **Owner:** you, the Superintendent
- **With:** your supervisors; the department's leads in Block 2
- **When:** 05:00 to 17:00, on the supervisors' clock. With several supervisors, see "More than three supervisors, and the night shift" in Step 2. Times are indicative.

Your four blocks, and what you check in each

The four blocks are fixed, and so are the touch-base and the check-in. Each bar below holds everything you check in that block.



Your four blocks across the day, with everything you check in each. Orange markers: where you meet your supervisors. Times are indicative.

3. The five stepping stones of your day

Block 1: check before the pit commits. At the handover, the pit plan and layout are checked against the mud map or pit map, not walked. Check that each supervisor has worked out their numbers (see "Checking the supervisor's numbers" in Step 2). After the pre-start, see the ground for yourself before anyone tells you about it. At the combined Drive-around, the supervisor tells you what they see and you tell them what you see.

Block 2: fixes back before the shift is lost. Take the committed numbers and what the supervisors cannot fix into the Daily Department Review (Step 3) and the Daily Site Review (Step 4). Test tomorrow's plan against today's ground. Leave office work for Block 3.

Block 3: what outlives this swing. People, training and the plan: the work that only you can do, with field evidence in hand.

Block 4: verify and coach on the ground. There is no check-in in Block 4. Make sure the pit goes hard until the last bucket. Agree any risk accepted into the next shift, such as a toe left or a face not prepped. At the final drive-around the supervisor explains the variances against the plan and their impact: the actions taken and whether they worked, what could have foreseen or avoided them, and whether they reach the next shift. You check; you do not explain for them.

Reading the Shift Report. The supervisor's Shift Report is two pages (the template is in Step 4 of the Supervisor Handbook; the blank is Apply Tool 6 in the supervisor's Apply tools). Before the final drive-around, check it has: the number committed at the Drive-around, written at the top; production by digger, in loads and buckets per hour, plan against actual, with payload and digging hours; each block's loads, with the system's figure beside the operators'; first and last bucket and cycle time; the delays; safety and the pit, including what we got away with; the gap in one line; each variance explained; jobs not finished and the risks to the next shift; the things escalated and still to be resolved; and the key actions needed from the next shift. If a line is missing, ask for it at the final drive-around, not after.

How your blocks link to your supervisors' blocks

Your four blocks are linked to your supervisors' four blocks, on the same clock (side by side, page 24).

When do you spend time with your supervisors?

At the end of each supervisor's Block 1 you have a combined Drive-around: they say what number they can achieve, and whether the handover they got and the assumptions made for the shift are accurate. Keep it to time: their block is theirs to run.

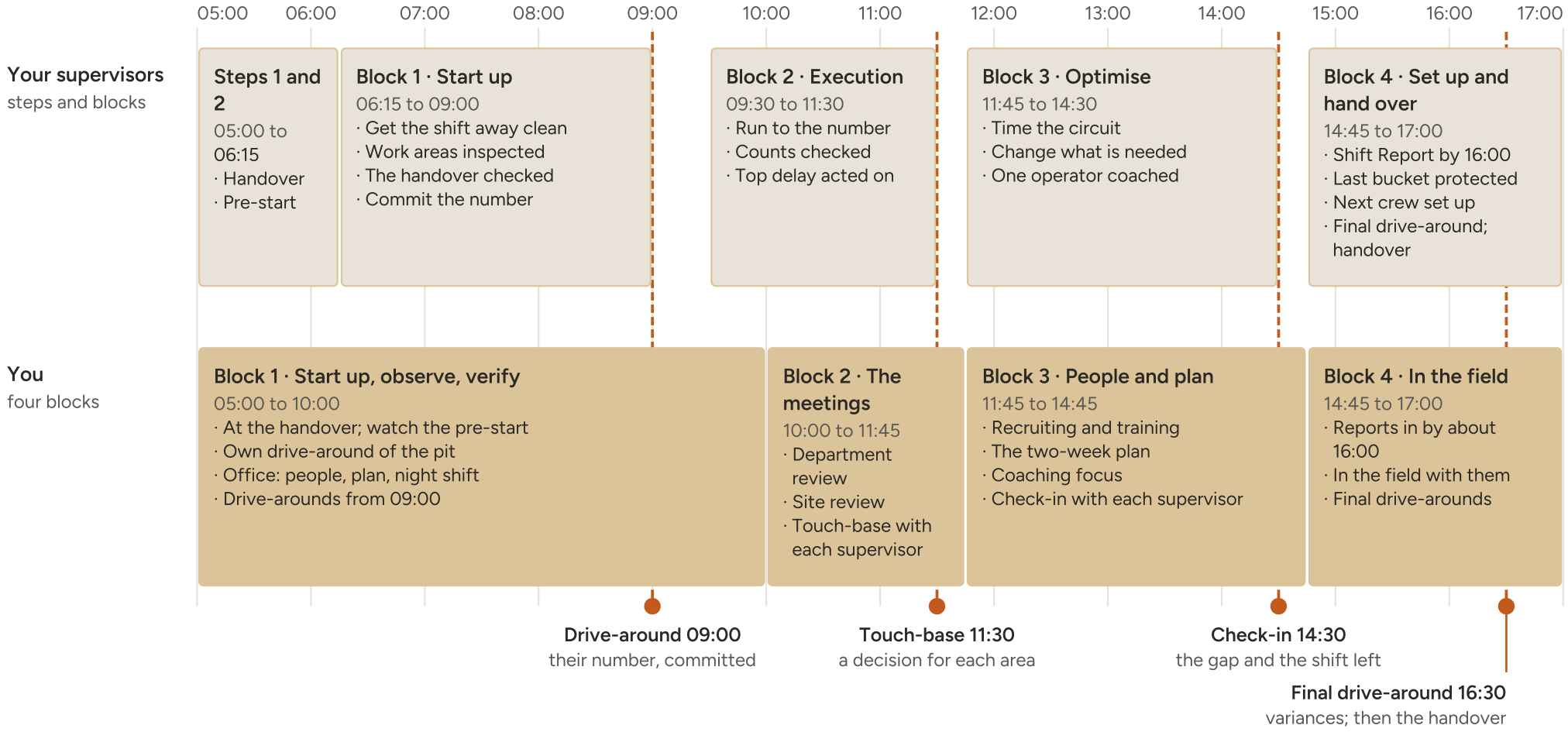
At 11:30, the touch-base, preferably at the lookout: their loads and buckets for the first part of the day, whether they will hit the number they committed to, and what they need to escalate. At 14:30, the check-in: the gap, and the shift left.

In Block 4, once their write-up is done, you are in the field together until the last bucket, then the final drive-around.

Your supervisors are not at the Daily Department Review, the Daily Site Review or the weekly and monthly reviews. They are in the pit. With several supervisors, you see each in turn: Drive-arounds from 09:00, touch-bases from 11:30, check-ins from 14:30, final drive-arounds from 16:30.

Your blocks and your supervisors' blocks, side by side

Your supervisors' day and your day on one clock: their steps and blocks on top, your blocks underneath. The orange lines are where you meet and something passes between you.



One clock. Orange lines: where you meet your supervisors and something passes between you. Times are indicative.

Coaching card: your four blocks

Yes / No

Stand-alone tool: your four block cards in Apply Tool 1. You mark each line Yes or No yourself at the end of its block.

#	What good looks like	Y/N
Block 1 · Start up, observe, verify		
1	Handover and pre-start watched without taking over, and coached from the supervisor's cards; then your own drive-around of the pit, before the office.	
2	People counted by crew and area.	
3	The plan and the next 24 to 72 hours checked; the contingencies still right.	
4	The night shift's work verified against what it committed.	
5	Every supervisor's Drive-around done at the end of their Block 1, before 10:00, by you or a named lead.	
Block 2 · The meetings		
6	Tomorrow's plan tested against today's ground, after the Daily Site Review.	
7	Back in the pit for the touch-base 11:30, preferably at the lookout; office work left for Block 3.	
Block 3 · People and plan		
8	Recruiting, training and the two-week plan moved on; a staff shortage ordered to hold face positions.	
9	Each supervisor's coaching focus for tomorrow written down.	
10	One decision a supervisor keeps making live placed before the work, written down.	
Block 4 · Time in the field, final drive-arounds		
11	In the field with each supervisor after their handover write-up, until the last bucket; the areas they inspected re-checked against the written standard.	
12	The work checked on the ground against what was committed; future shifts walked: blasted stocks, benches, floors and dumps.	
13	The night shift's crew, gear and plan confirmed before you leave.	
14	Last final drive-around's coaching point checked with each supervisor: done or not.	
+	Safe to speak up: people asked, challenged and admitted mistakes without being put down.	

Actions required from any No: ___ Owner: ___ By: ___

Worked example: a coaching focus and a rule placed (Tool 1)

Supervisor	The No	Coaching focus for tomorrow	Owner	By
A crew	Line 8: the maintenance row ticked without the plan in hand	Read the maintenance plan at 05:00, before the cross-process check	A crew supervisor	Next shift's handover

Block 3, line 10: placed before the work. A crew's supervisor put forward: "A truck service is booked before the shift change, so neither shift loses digging hours." You agreed it with the Maintenance Superintendent at the next Daily Site Review and wrote it in the commitments register. From then on it is checked at Block 1, not decided at the Drive-around.

When you do this, use: Apply Tool 1, your four block cards.

Step 2 The Drive-around cards

Key question: What number is each supervisor committing to, having seen the ground, and what do they need from me to hold it?

The Drive-around, the touch-base and the check-in are where you meet each supervisor in the shift. At 09:00 you drive the pit together and the supervisor commits a number, having seen the ground. At the 11:30 touch-base and the 14:30 check-in you hear the block in numbers and each area leaves with a decision. Done well, you know of every major variance before any report tells you, and no supervisor leaves the pit to get a fix.

Outcome: Each supervisor's number committed on the ground and tested, the fixes taken back to the pit, and every variance known by the check-in.

- **Owner:** you, the Superintendent
- **With:** each supervisor, one at a time
- **When:** Drive-around from 09:00, at the end of the supervisor's Block 1, about 30 minutes with one supervisor; touch-base from 11:30 (preferably at the lookout) and check-in from 14:30, 15 minutes each, face to face. Day shift only. Times are indicative.

Decisions the Drive-around, touch-base and check-in must make (tick each)

#	Decision	Done
1	The number for the shift: accepted, or an input changed	☐
2	For a short crew or a unit down: the priority for the rest of the shift	☐
3	What goes up to the Daily Department Review	☐
4	At the touch-base and the check-in, for each area: continue, correct, prepare or escalate	☐

The Drive-around, and the question asked

Min	Item	Lead	The question asked
5	The ground, seen	Supervisor	What did you find when you checked the handover?
5	Rules and contingencies, out loud	Supervisor	If the digger breaks down, where do the trucks go?
10	The number, committed	Supervisor	What number are you committing to, in loads, and how many loads and buckets an hour is that?
5	Below target	Superintendent	What on the ground is the reason, and which assumption has moved?
5	What goes up	Supervisor	What can you not fix yourself? What do you need from me?

The touch-base and the check-in: four questions, in numbers

1. Loads, buckets, payloads and delays this block, from the operators' counts, against target?
2. What changed, and did it work?
3. What do you need from me?
4. What do you escalate?

At the 11:30 touch-base you hear the loads and buckets per hour and the payloads, and the top delay and what was done about it. At the 14:30 check-in you hear the stopwatch, the gap and how much shift is left, and the one change made.

Rule: from the Drive-around on, the supervisor delivers against a number they own. You test a number below target; you do not overrule it. A supervisor who commits below target must be clear why, and must name what needs fixing that they cannot fix themselves.

Checking the supervisor's numbers

Each supervisor works the target out at the handover, from their own machines, this face and this material, and says it at the pre-start (Supervisor Handbook, "Know your numbers"). The number they commit is in loads; they report it in loads and buckets per hour. Buckets a minute and seconds a bucket are only for checking at the face. At the handover, check the working; at the Drive-around, test the committed number against it.

Step	The calculation	Worked example
1 · Shift volume	From the weekly plan, adjusted for today's machines: bcm or t	12,000 bcm
2 · Loads	Volume ÷ payload per load = loads	12,000 ÷ 60 = 200 loads
3 · Digging hours	Shift hours − pre-start − travel − crib − fuel − shift change = digging hours	12 − 3 (pre-start and travel, crib, fuel, shift change) = 9 h
4 · Loads an hour	Loads ÷ digging hours = loads an hour; ÷ trucks = per truck per hour	200 ÷ 9 = about 22 an hour; ÷ 4 trucks = about 5.5 per truck per hour
5 · Buckets a load	Buckets that fill a truck on this face, with this machine, in this material	5
6 · Buckets a minute, seconds a bucket	Loads an hour × buckets a load ÷ 60 = buckets a minute; 60 ÷ buckets a minute = seconds a bucket	22.2 × 5 ÷ 60 = about 1.85 a minute; 60 ÷ 1.85 = about 32 s

Ask the supervisor to say it in one sentence: "We need ___ loads this shift: that is ___ an hour, ___ per truck per hour, and a bucket every ___ seconds." If they cannot, the number is reported, not committed. Where the payload or digging hours differ from what the site was priced on, that is a finding for Apply Tool 11.

Coaching card: the Drive-around cards

Yes / No

Card 1, the Drive-around: lines 1 to 7. Card 2, the touch-base and the check-in: lines 8 to 10.

Stand-alone tools: Drive-around card 1 is Apply Tool 2; card 2 is Tool 3.

#	What good looks like	Y/N
1	The pit driven with the supervisor at the end of their Block 1, in one vehicle: the ground seen, not asked about.	
2	Golden rules, TARPs and the shift's contingencies said out loud.	
3	Short of crew, or a unit down: the priority for the rest of the shift agreed.	
4	A number below target tested, not overruled: an input changed or the number accepted, never the target simply insisted on.	
5	Every risk owned by a name, or escalated the same morning.	
6	What the supervisors cannot fix taken to the Daily Department Review, with the department total.	
7	The fix or contingency taken back to the supervisor in the pit, in Block 2 or at the 11:30 touch-base: the supervisor does not leave the pit.	
8	The touch-base and the check-in heard in numbers: loads, buckets, payloads and delays against target.	
9	Each ends in a decision for each work area: continue, correct, prepare or escalate.	
10	Major variance learned at the touch-base or the check-in, before the report tells you.	
+	Safe to speak up: people asked, challenged and admitted mistakes without being put down.	

Actions required from any No: ___ Owner: ___ By: ___

More than three supervisors, and the night shift

- **Several supervisors.** You see each in turn. The Drive-arounds are staggered from 09:00, and each supervisor's Block 1 runs on until theirs. The touch-bases are staggered from 11:30 and the check-ins from 14:30, and each supervisor's next block starts after their own. Final drive-arounds, after the last bucket: one supervisor, 16:30 to 16:50; two, 15 minutes each, at 16:30 and 16:45; three, 10 minutes each, at 16:30, 16:40 and 16:50. All end by 17:00. Each report is done by about 16:00. Times are indicative.
- **More than three supervisors.** You drive with three at most: about 30 minutes each for two, about 20 minutes each for three, so the Drive-arounds end by 10:00. Each further group of up to three has a named lead who runs its Drive-arounds, touch-bases and check-ins on the same cards, and gives you the numbers before the Daily Department Review. Rotate through the groups so every supervisor has a Drive-around with you at least once a week.
- **The 20-minute Drive-around (three supervisors).** 5 minutes, the ground seen; 3 minutes, rules and contingencies said out loud; 7 minutes, the number committed, and tested if below target; 5 minutes, what goes up.
- **Night shift.** No Drive-around, no touch-base and no check-in. The supervisor writes the number at the top of the Shift Report and escalates by radio or phone when recovery is beyond the shift. At the start of the morning handover the night supervisor explains the variances from the night block cards, No answers first (their final drive-around); then the supervisors run the handover.

Worked example: Drive-around card 1, A crew (Tool 2)

Field	Entry
Block 1 loads: operators' counts / the system	47 / 46 (one load with no payload reading)
The number committed	190 loads; about 21 loads an hour over 9 digging hours; about 105 buckets an hour
Below target because	T3 booked for a 2-hour service at 12:00: 10 loads
Tested	Asked what number with four trucks all shift: 200. Accepted 190 for now; the service time goes up.
What goes up	T3's service time: needs Maintenance

When you do this, use: Apply Tool 2, Drive-around card 1, and Tool 3, Drive-around card 2.

Step 3 The Daily Department Review

Key question: What does each supervisor need back before 11:30, and what can only the site settle?

The Daily Department Review is where your department settles what its supervisors cannot fix, before the shift is lost. You chair it at 10:00, straight after the Drive-arounds, with the department's leads. Done well, it takes 20 minutes, every unfixable leaves with a fix or a contingency on its way back to the pit before the 11:30 touch-base, and only what needs another department goes on to the Daily Site Review.

Outcome: A fix or contingency for every issue a supervisor cannot fix, on its way to the pit before 11:30, and the department's line for the Daily Site Review.

- **Owner:** you, the Superintendent
- **Attends:** the department's leads: planning, drill and blast, the maintenance planner for the fleet, and training; Technical Services when needed
- **When:** daily, 10:00, 20 minutes. Supervisors do not attend; they are in the pit. Times are indicative.
- **Feeds:** the supervisors (fixes before their 11:30 touch-base) and the Daily Site Review at 10:30

Decisions this meeting must make (tick each)

#	Decision	Done
1	Each issue a supervisor cannot fix: the fix or contingency, an owner and a time	☐
2	What the department settles itself, and what needs another department	☐
3	The department's line for the Daily Site Review: the number, the gap and its reason, and the asks	☐
4	Today's department priorities, in order	☐

What to bring and the agenda

Bring to the meeting

What	Who
Each supervisor's committed number and gap, and what they cannot fix	You, from the Drive-arounds
Yesterday's shift reports, in PAVA	You, from the final drive-arounds
Today's drill, blast and maintenance plans for the department	The leads
The department log: yesterday's actions	You

Agenda (20 minutes)

Min	Item	Lead	Output
2	Yesterday's actions in the log: done or not	Chair	Log updated
5	Each supervisor's number and gap	Chair	The department total
8	Each issue a supervisor cannot fix	Chair	A fix or contingency, an owner and a time
3	The line for the Daily Site Review	Chair	The number, the gap, the asks
2	Close: who takes each fix to the pit	Chair	Fixes on their way

Rule: a cause that comes back for the third time is not fixed again here. It goes on the department page for the Weekly Site Review, with the evidence.

Coaching card: the Daily Department Review

Yes / No

Stand-alone tool: the Daily Department Review card, with the log in Apply Tool 5.

#	What good looks like	Y/N
1	Starts at 10:00 and ends inside 20 minutes.	
2	The department log read first: yesterday's actions done or not.	
3	Every supervisor's committed number and gap on the table, from the Drive-arounds.	
4	Each issue a supervisor cannot fix leaves with a fix or contingency, an owner and a time.	
5	Every fix leaves with the name of who takes it to the pit.	
6	Yesterday's variances read in PAVA, from the shift reports.	
7	A cause seen for the third time flagged for the department page.	
8	What the department can settle is settled here; only what needs another department goes to 10:30.	
9	The department's line for the Daily Site Review agreed: the number, the gap and its reason, and the asks.	
10	Today's department priorities leave in order.	
+	Safe to speak up: people asked, challenged and admitted mistakes without being put down.	

Actions required from any No: ___ Owner: ___ By: ___

Worked example: the department log (Tool 5)

Supervisor	Committed / target	What they cannot fix	Fix or contingency	Owner	By	Settled here, or to 10:30
A crew, EX1	190 / 200	T3 booked for service at 12:00	Needs Maintenance to move it	You	10:30	To the Daily Site Review
EX2 circuit	196 / 200	No trainer for the new truck operator	Trainer moved from the dump crew for the morning	Trainer	11:00	Settled here

The line for the Daily Site Review: "Load and haul: committed 386 of 400 loads. Gap 14: T3's service at 12:00 (10) and a new operator on EX2 (4). Ask: T3's service moved out of digging hours."

When you do this, use: Apply Tool 5, the Daily Department Review log.

Step 4 The Daily Site Review (you chair it)

Key question: Where do the departments' plans collide today, and what do we settle so that the committed numbers hold?

The Daily Site Review is where the departments settle what one department cannot settle alone. You chair it at 10:30, after the Daily Department Reviews. The Project Manager attends, with the Superintendents of every department, Maintenance, Safety and HR. Done well, the room decides between departments and does not review again within one; the site leaves with one priority list for the next 24 hours; and what cannot be fixed goes back the same shift. Supervisors do not attend; they are running the pit.

Outcome: The collisions between departments settled, one priority list for the next 24 hours, and every fix on its way back the same shift.

- **Owner:** you, the Superintendent (where a site has several, it names the one who chairs)
- **Attends:** the Project Manager; the Superintendents of every department; Maintenance, Safety and HR
- **When:** daily, 10:30, 30 minutes. Times are indicative.
- **Feeds:** the supervisors (fixes in Block 2 or at the 11:30 touch-base) and the Weekly Site Review (repeat collisions)

Decisions this meeting must make (tick each)

#	Decision	Done
1	Each collision between departments (gear, sequence or priority), settled today	☐
2	Each department's ask: a fix or contingency, an owner and a time inside this shift	☐
3	The site's next 24 hours: one priority list every department plans against	☐
4	Where management help goes, chosen by impact	☐
5	What leaves site level, with an owner and a forum	☐

What to bring and the agenda

Bring to the meeting

What	Who
Each department's line: the number, the gap and its reason, the asks	Each Superintendent, from their Daily Department Review
Today's maintenance plan, and units due back	Maintenance
The day's safety signals	Safety
People: absences and who is short	HR
The site decision log: yesterday's decisions	You

Agenda (30 minutes)

Min	Item	Lead	Output
5	Safety across the site: the day's signals, as a pattern	Chair	One signal looked at properly
10	Each department in one minute: the number, the gap, the asks	Superintendents	The site's day on the table
8	The collisions and the asks, decided	Chair	Each with an owner and a time
5	The next 24 hours; where management help goes	Chair	One priority list
2	Close: what goes back this shift, and by whom	Chair	Decisions logged

Rule: a collision that comes back every week is a fault in how the work is set up. It goes to the Weekly Site Review as a system finding, not back to the departments to settle again each day. What the heads of department cannot settle goes to the Project Manager, with the number it protects.

Coaching card: the Daily Site Review (you chair it)

Yes / No

Stand-alone tool: the Daily Site Review card, with the log in Apply Tool 6.

#	What good looks like	Y/N
1	Starts at 10:30 and ends inside 30 minutes; the Superintendent chairs and the Project Manager attends.	
2	Safety signals read as a pattern first: one looked at properly.	
3	Each department in one minute: its number, the gap and its reason, and what it cannot fix.	
4	The room decides between departments; nothing is reviewed again within one.	
5	Every clash between departments settled today, with an owner.	
6	What cannot be fixed goes back the same shift, with a contingency.	
7	The site's next 24 hours leave as one priority list, said once.	
8	Where management help goes today is named, chosen by impact.	
9	What the site cannot settle goes up with an owner and a forum.	
10	The site decision log filled in, one row per process.	
+	Safe to speak up: people asked, challenged and admitted mistakes without being put down.	

Actions required from any No: ___ Owner: ___ By: ___

Worked example: the site decision log (Tool 6)

Process	Issue or collision	Decision	Owner	By	Back to
Drill	Rig cannot start bench 4 until its floor is cleaned up	A crew's dozer on bench 4 from 12:00, to be agreed with the A crew supervisor at 11:30	You	12:00	A crew supervisor, at 11:30
Blast	Bench 3 shot at 16:45	Clearing times to every crew	Drill and blast Superintendent	14:00	All supervisors
Load and haul	T3 service at 12:00 costs 10 loads	Service moved to 16:30, before the shift change	Maintenance Superintendent	11:00	A crew supervisor, at 11:30
ROM	Pad 2 full by 15:00	Pad 3 opened from 14:00	Processing Superintendent	14:00	Load and haul supervisors
Maintenance	EX1 250-hour service due next week	To the Weekly Site Review as a threat to next week	Maintenance Superintendent	Weekly Site Review	Project Manager

Help today goes to EX2's new operator: a trainer for the morning.

When you do this, use: Apply Tool 6, the Daily Site Review log.

Step 5 Taking part in the Weekly Site Review

Key question: Where did my department's week land, and what is coming that could impact next week?

The Weekly Site Review is the site's weekly IN meeting, in weeks 1 to 3. The Project Manager chairs it; you attend for your department. It is operational only: task slips, issues and workloads, and next week's threats, each given to a name. Done well, your department page is in the day before, your two minutes are numbers, not stories, and you leave with every threat to next week owned.

Outcome: Your department's week judged in two minutes, its top delays owned, and every threat to next week given to a name.

- **Owner:** the Project Manager
- **Attends:** you and the other Superintendents, Department Managers, Maintenance, HSEC, HR, Technical Services
- **When:** weekly, weeks 1 to 3, 60 minutes. In week 4 it becomes the first hour of the Monthly Site Review.
- **Feeds:** the Weekly Site Planning Meeting, where next week's plan is made, and the head office IN meeting

Decisions you take part in (tick each)

#	Decision	Done
1	Each of your department's task slips and issues: recovered, with an owner and a date	☐
2	Each threat to next week: given to a name, with what they will do and by when	☐
3	What changes in next week's plan, roster or maintenance windows, for the Weekly Site Planning Meeting	☐
4	The site's priorities: one list	☐
5	What goes up from your department, with an owner and a forum	☐

What to bring and the agenda

Bring to the meeting: your department page (the day before)

What	Who
The week in PAVA: loads and the measures that moved	You
The top 3 delays, each with root cause, fix, owner and date	You
Next week's threats: weather, crews, maintenance, the plan	You
Repeat causes (seen three times), with the evidence	You

Build the page from the week's department logs, shift reports and final drive-arounds. For each of the top 3 delays, walk the fault tree to the root cause (chapter 4).

Agenda (60 minutes)

Min	Item	Lead	Output
5	Last week's actions: done or not, and why	Chair	Register updated
5	Weak signals: what are we repeatedly getting away with?	Chair	The pattern named
15	Each department's week, two minutes each: task slips, issues and top delays	Superintendents	What each department needs
10	Workloads: who is over and who has room	Chair	Workloads rebalanced
15	What is coming: weather, crews, recruitment, maintenance	Chair	Each threat given to a name
10	Close: the site's priorities; what goes up	Chair	One list; owner and forum for each item

Coaching card: the Weekly Site Review

Yes / No

Stand-alone tool: your Weekly Site Review card, with the department page in Apply Tool 7.

#	What good looks like	Y/N
1	Department page in to the Project Manager the day before.	
2	Your week said in two minutes, in PAVA: task slips and issues, not stories.	
3	Each of the top 3 delays has a root cause, a fix, an owner and a date.	
4	Each threat to next week given to a name before the meeting ends.	
5	Changes to next week's plan, roster or maintenance windows taken to the Weekly Site Planning Meeting.	
6	Last week's actions for your department reported: done or not, and why.	
7	Weak signals named: what your department keeps getting away with.	
8	Your department's workload said: where it is over, and where it has room.	
9	Repeat causes (seen three times) brought with the evidence, as system findings.	
10	You leave with the site's priorities as one list, and each item from your department has an owner and a forum.	
+	Safe to speak up: people asked, challenged and admitted mistakes without being put down.	

Actions required from any No: ___ Owner: ___ By: ___

Once a month. In week 4 the Weekly Site Review becomes the first hour of the Monthly Site Review, the site's ON meeting, also chaired by the Project Manager. There you bring the coaching card records for the system being examined.

Worked example: the department page, North Pit, load and haul (Tool 7)

Measure	Plan	Actual	Variance	Action
Loads, the week (14 shifts)	5,600	5,380	−220 (−4%)	Top 3 delays below
First bucket, average	06:30	06:38	8 min late	Fuel truck to the diggers by 06:00: owner, the supervisors, from Monday

#	Delay	Loads lost	Root cause	Fix	Owner	By
1	Truck services in digging hours	120	Services booked by the maintenance planner without the digging plan	Rule: services booked before the shift change, so neither shift loses digging hours	Maintenance Superintendent	Done
2	Late first bucket	60	Fuel truck reaches the diggers at 06:20	Fuel by 06:00	Supervisors	Monday
3	Trucks waiting at the face	40	Five trucks on EX2 when the haul shortened	Four trucks on EX2 while the haul is under 2 km	EX2 supervisor	Monday

When you do this, use: Apply Tool 7, the weekly department page.

Next week's threats: rain forecast for two days (owner: you, wet-weather roads plan); two operators on leave (owner: you, roster); EX1's 250-hour service (owner: Maintenance, on night shift).



4. Diagnostics

When the number is at risk, find out why, and fix the cause, not the symptom.

Section 1 · Reading the mine

Section 2 · How your performance is measured: PADAL

Section 3 · Productivity levers

Section 4 · Finding the cause

Section 5 · The fault trees

Reading the mine

The department is one circuit across processes. Drill, blast, load and haul, ROM and maintenance each hand work to the next. A blast that fires in digging hours, a service booked at the wrong time or a ROM pad that fills early shows up as lost loads in someone else's circuit. Your supervisors see their circuit. You see how the processes meet, and settle where they collide.

Know your contract. How the contract pays decides what counts. If it is paid on rates and volume, the money is in the volume moved, and the plant is charged for every hour the engine runs. Revenue is made only in primary work time: the hours the digger is loading. An engine hour that moves no volume is paid for and earns nothing. If it is paid on SMU hours only, it is usually just a staff rate and an hours rate. Most contracts sit somewhere in the middle. Part B of this program covers this in more detail, but you and your supervisors should know your site's contract now, because the PADAL measures you are judged on may change with it. Tell your supervisors which of the five your site reports most. Load and haul is a large share of a mine's operating cost, and an hour lost on a digger takes the whole circuit behind it with it.

How your performance is measured: PADAL

Your department is measured on five numbers, called PADAL. Every one of them moves on your supervisors' shifts. PAVA is how you report; PADAL is what you are measured on.

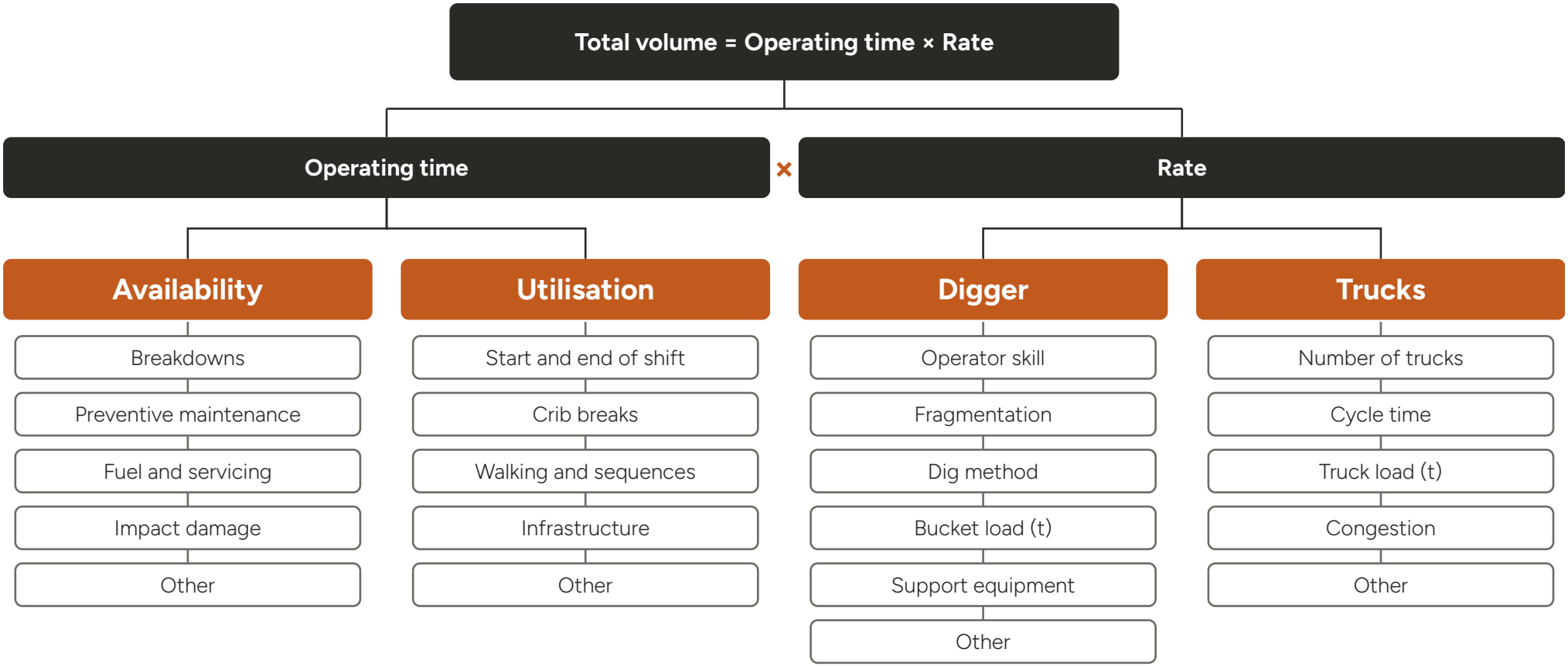
Letter	The number	In plain words	How it is worked out	Where the shift moves it
P	Primary productivity	How much is moved for each hour the digger is loading. Higher is better.	Output ÷ primary work hours	First and last bucket; hang time; buckets a minute
A	Annualised work hours	How many hours the gear works, scaled up to a full year. Higher is better.	(Work time ÷ calendar time) × 8,760	Every hour of delay, queue or waiting
D	Delay SMU percentage	The share of engine hours lost to delay instead of work. Lower is better.	(Engine on − work time) ÷ engine on	Waiting for trucks, fuel, a release, the blast
A	Ancillary factors	Hours of support gear (dozers, graders, water carts) for every 1,000 bcm or tonnes moved. Lower is better.	(Ancillary work hours × 1,000) ÷ output	Support gear on task, to plan
L	Labour factor	People hours for each machine engine hour. Lower is better.	Employee hours ÷ plant SMU hours	Seats manned against machines; trainers accounted for

When a supervisor commits under target at the Drive-around, ask which PADAL number moved and by how much. "D is up because we waited 40 minutes on the release" can be fixed. "Maintenance let us down" cannot. When a number is at risk, the rest of this chapter shows how to find out why.

Productivity levers

Key question: Which of these levers could or should be pulled to increase productivity around your operation?

In summary, there are only two levers that move total volume: operating time and rate. Your supervisors pull them inside the shift. You pull them across shifts and across processes, and you fix what keeps a lever stuck. This is the first stop when you reflect on the day, the week or the department page: start here.



Reading the levers

Operating time is availability × utilisation. Rate is what the digger and the trucks do with the time they get. Two more causes sit under the rate on shift: the floor the digger works on, and the trucks' spot and queue time at the face.

Every delay your supervisors code, and every rate they lose, lands on one of these levers. Name the lever before you name the fix. On the department page, quantify the loss, locate the lever, then raise the action or the project. "We need to lift productivity" is a wish. "We lose 40 minutes a shift to start-of-shift utilisation, and here are the three causes" is a decision waiting to be made. The rest of this chapter, and the fault trees at the end of it, help you find the cause behind the lever.

Finding the cause

Your supervisors diagnose the shift, in the field, above all in Block 3 (Supervisor Handbook chapter 4). You diagnose what keeps coming back, and you coach them to look for the quality of the work, not just tasks done. A busy pit can still be losing the number. Use this chapter at the Drive-around, the touch-base and the check-in to test a number, in Block 4 to check the work on the ground, and on the department page to find the root cause of the top 3 delays. First name the productivity lever, then look at the circuit, the dig and the dump, the truck allocation and the loading.

Which part of management failed? Every fix falls under define, allocate, coordinate or verify (the definition of management, page 8). When something goes wrong, ask which one failed.

Checking the circuit, the dig and the dump

Circuit efficiency. Your supervisors time the circuit in Block 3 (line 30) and bring it to the 14:30 check-in. In Block 4, check it yourself: stand where you can see the whole pit, then put a stopwatch on three trucks, not one:

- **Hang time:** the digger waiting, bucket full, for a truck to spot.
- **Queue time:** a truck waiting at the face for the one in front.
- **Load time:** spot plus load, truck stopped to last bucket.
- **Cycle time:** face to dump to face.

Compare them with the target cycle for this haul. Hang and queue together tell you whether the circuit is short of trucks or has too many. Load time tells you about the face and the operator. Cycle time tells you about the road. Industry studies suggest a 10% rise in waiting time costs a few per cent of actual productivity (indicative). The digger operator sees the circuit from the face: ask whether your supervisor asked them.

The dig set-up: the loading face. Good looks like: the digger loads without waiting, trucks arrive already in position, and the payload sits in its band. The circuit is usually lost in the last 50 metres. Loading from both sides (double-side loading) removes truck exchange and is often credited with a gain of a few per cent over loading from one side (indicative). Each supervisor sets the bucket rate from their own machines. Check:

- bucket-to-bucket time: the gap between one bucket landing and the next;
- spot time: how long a truck takes to get into position;
- the floor, the toe and the face position, and bench height against the machine.

The dump set-up: the dump and tip head. Good looks like: trucks tip without hesitation, the edge is to standard every round, and the order of tipping matches the plan. Check:

- the tip-head edge and windrow: to standard, and lit at night;
- the dump level and design width: survey confirming the room left;
- the dozer on task: pushing to design, not idling;
- the dump sequence against the plan: cells used in the order planned.

Haul roads and intersections. Good looks like: trucks hold a steady speed without hunting for gears, the surface is kept up before it breaks down, and the grader works to a plan. Haulage can be more than half of the operating cost in a deep pit. Industry studies suggest that managing road condition and grade together cuts cycle time and fuel, and lifts each truck's output (indicative; it varies by site). Check the surface (corrugations, potholes, soft spots, loose rock), the width, grade and camber, the intersections (sight lines, signs, who has priority), and where trucks change gear: a grade that makes them hunt is costing you.

Truck allocation. Match the trucks to the circuit. Too many trucks slow a circuit and burn fuel: queue time goes up. Too few leave the digger hanging. Do you have slower or smaller trucks in the circuit? Pulling them out can improve the speed and efficiency of the circuit. Sometimes sending an operator to crib costs less than manning a truck you do not need.

Loading: are your trucks loaded properly? How do you know? Check the payload against the target for each truck: under-loading wastes the circuit, and over-loading wastes the truck. Count the passes (buckets) a load on this face. If it takes more passes than the target, find out why: bucket fill, fragmentation or the operator. Where the site has payload readings, put them beside the operators' counts. Check the payload against what the circuit was priced on (chapter 2, "The assumptions the site was priced on"; Apply Tool 11).

The enablers

None of these moves a tonne on its own. All of them decide whether the circuit can.

- **Water and pumps:** the next face is dry before the plan reaches it, and the wet-weather fallback is named and known. Test it at every Daily Department Review.
- **Drill and blast:** enough broken rock in front of the diggers that a hard band or a late blast does not stop digging. Test it at every Daily Department Review, against the two-week plan.
- **Support gear, fuel and the go-line:** dozers, graders and water carts where the plan needs them, before it needs them; fuel in its window; no unit parked without somebody knowing why. This one is the supervisor's: check that they own it.

The fault trees

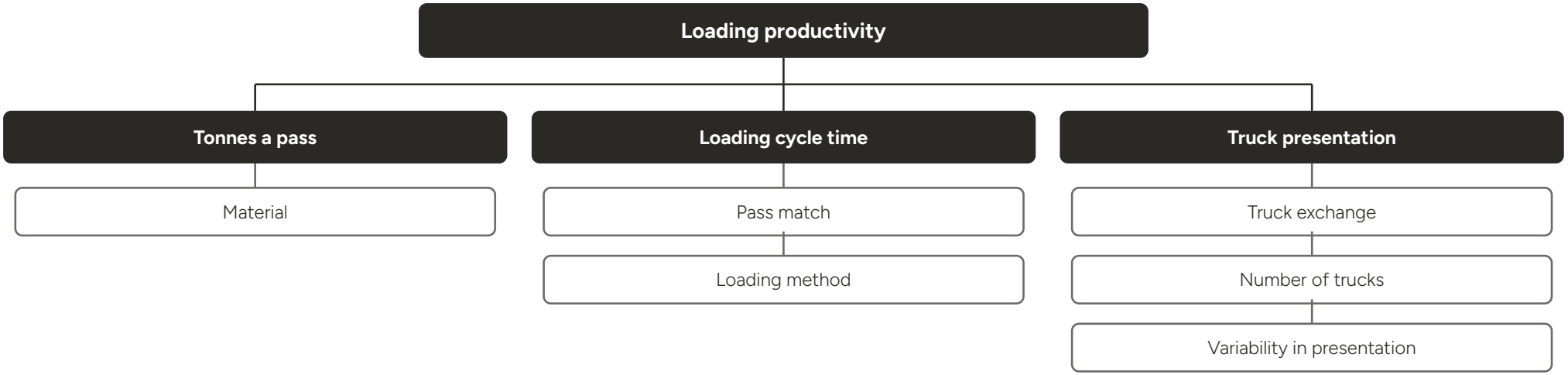
The fault trees support the productivity levers: they take a lever down to its cause. Fault tree 1 sits under the rate (the digger and the trucks); fault tree 2 sits under operating time (availability and utilisation). When a loss keeps coming back, walk the fault tree. Start at the top and work down. Do not stop at the branch: the branch is the symptom. Keep going until you reach a box you can change, then check the foundation bar, because most branches end there.

How to use it: take the single biggest loss of the week and walk the tree out loud with the supervisor, or at the Daily Department Review, naming the box at each level. Where two people name different boxes, you have found a data problem, not a disagreement. Write down the lowest box you reach as the cause. If you land on the foundation bar, say so plainly; do not call it an equipment fault.

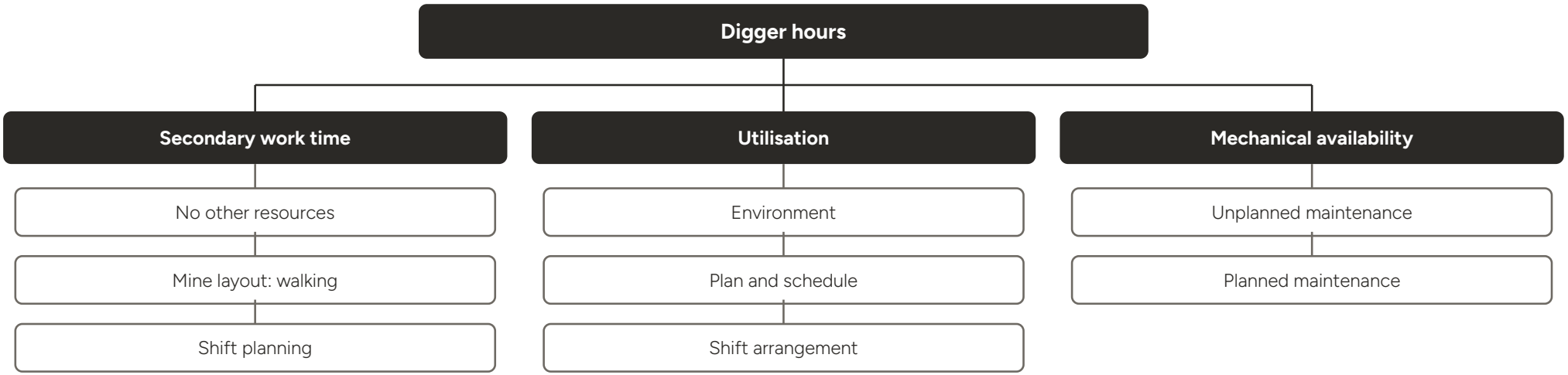
The full trees, every branch to its root, are the A4 fault-tree tool (the supervisors' Apply Tool 9). Put what you find on the department page for the Weekly Site Review (Apply Tool 7).

Underneath every branch: the foundation bar. Management · Standards and procedures · Roles and decision rights · Competence and training · Supervision in the field · Culture.

Fault tree 1: loading productivity



Fault tree 2: digger hours (primary work time)



Fault tree 1, branch by branch

Branch	Second level	Third level, and what sits under it
Tonnes a pass	Material	How loose the muckpile is: design · drilling · shotfiring
Loading cycle time	Pass match (number of passes)	Bucket: maker's standards · other buckets available. Truck tray: payload limits · other trays available
Loading cycle time	Loading method	Planning (week and beyond): design · geology · objectives. Work area: shift planning · standards · operator competence. Material: several types
Truck presentation	Truck exchange	Work area: planning · competence · supervision. Competence: training · supervision
Truck presentation	Number of trucks	Shift planning: week planning · supervision. Availability: planned · unplanned
Truck presentation	Variability in presentation	Number of trucks: supervision · mechanical availability · operator. Different destinations: planning · material types · supervision. Haul road interference: planning · supervision. Competence: training · supervision

Fault tree 2, branch by branch

Branch	Second level	Third level, and what sits under it
Secondary work time	No other resources	Shift planning: supervision · weekly planning
Secondary work time	Mine layout (walking)	Design · standards and procedures
Secondary work time	Shift planning	Supervision
Utilisation	Environment	Wet. Wet recovery: haul road building · supervision. Fog and lightning: standards and procedures · geography
Utilisation	Plan and schedule	Go straight to the foundation bar. Was the plan made before the work, and was it followed?
Utilisation	Shift arrangement	Blast delays: shift planning · week planning. Crib delays: supervision · standards · data to track
Mechanical availability	Unplanned maintenance	Accident damage: work area · how production works · how maintenance works · environment. Routine maintenance missed: defects · machine housekeeping · short of skilled labour. Stores and supply: warehouse space · critical spares planning. Labour productivity: skills
Mechanical availability	Planned maintenance	Maintenance planning: standards · sequencing. Labour productivity: skills · supervision · taken off to unplanned work. Component life: target lives · operating conditions. Age of equipment: policy · replacement capital



5. Terms and academic models

Section 1 · Terms, A to N

Section 2 · Terms, O to Z

Section 3 · The evidence behind the model

Terms, A to N

Actions page: the page of the Shift Report where the actions from the final drive-around are written, each with an owner and a date.

Ancillary: support gear: dozers, graders and water carts.

Availability: the share of time the gear is fit to work: not broken down, and not in for service or fuel.

bcm: bank cubic metres: rock measured in the ground, before it is blasted.

Blasted stocks: rock drilled and blasted, ready to dig, in days of digging.

Block card: the coaching card for one block, marked Yes or No at the end of the block.

Block: one of the four parts of the day, two to three hours each, with its own card. The supervisor's Blocks 1 to 3 end with a meeting with you.

Check-in: 15 minutes with each supervisor, face to face, at 14:30 on day shift, in numbers. The 11:30 one is the touch-base.

Coaching card: Yes or No lines that say what good looks like. A No is the next thing to coach. Each card is a stand-alone tool, taken on shift.

Commitments register: the written list of rules and contingencies in force.

Contingency: an if/then fallback decided before the work, with a name on it.

Continue, correct, prepare, escalate: the four decisions each block, touch-base and check-in ends in.

Crib: the meal break.

Cycle time: a truck's round trip: queue, spot and load, haul, dump, and back to the face.

Daily Department Review: your department's 20-minute meeting at 10:00, where the supervisors' unfixables get a fix.

Daily Site Review: the 30-minute meeting of all departments at 10:30, which you chair.

Department page: one page for the Weekly Site Review: the week in PAVA, the top 3 delays, next week's threats.

Double-side loading: the digger loads trucks on both sides, so one truck spots while the other is being loaded.

Drive-around: you and a supervisor driving the pit together at the end of their Block 1, after your own drive-around, where the number is committed.

Face positions: where each digger will be digging over the coming weeks; falling behind them costs the plan.

Fault tree: a diagram that breaks a loss into its causes, branch by branch, down to a cause you can change.

Final drive-around: the last meeting with each supervisor, in Block 4, where the supervisor explains the shift's variances (the debrief).

Four core components of management: define the work, allocate it, coordinate it and verify it.

Go-line: where the gear is parked between shifts, ready to start.

Golden rules (guard rails): the site's standing rules, never decided again by whoever is on duty. They grow as decisions made live become rules.

Handover sheet: the sheet in the supervisor's Apply Tool 4 the off-going supervisor fills in for the oncoming one.

Hang time: the digger waiting, bucket full, for a truck.

High-reliability organisation (HRO): an organisation that does dangerous work every day with very few serious failures, because of how its people think and act on every shift.

HR: human resources.

HSEC: health, safety, environment and community.

IN and ON: IN meetings run the business (the week). ON meetings improve it (the month).

Level of work: the time span of the longest task a role is accountable for (Jaques; Systems Leadership). Supervisor II: 3 months to 1 year. Superintendent III: 1 to 2 years. Project Manager IV. On shift, your focus is the day across crews, the week and the month.

Load time: spot plus load, truck stopped to last bucket.

Lookout: a point above the pit where you can see the whole circuit; the touch-base is held there.

Manager once removed: the manager of a person's manager: the Superintendent, to an operator.

Monthly Site Review: the site's monthly ON meeting, in week 4, chaired by the Project Manager.

Mud map: a simple drawing of the pit for the shift: where each digger is, the roads, the dumps and the hazards.

Named lead: with more than three supervisors, the person who runs a group's Drive-arounds, touch-bases and check-ins on the same cards.

Number (the): the loads a supervisor commits to for the shift, at the Drive-around after Block 1, having seen the ground. Reported in loads and buckets per hour.

Terms, O to Z

Operator scorecard: each operator's trend over the roster, on the wall in the pre-start room.

PADAL: the five production measures: primary productivity, annualised work hours, delay SMU percentage, ancillary factors, labour factor.

Part B: the second part of this program, after Part A (this book). It covers optimal mining conditions, and how your contract pays, in more detail.

Pass: one bucket into a truck. Pass match: the number of passes it takes to fill the truck.

PAVA: Plan, Actual, Variance/Impact, Actions taken: how every number is shown.

Pit map: a map of the pit showing the faces, benches, roads and dumps. With the mud map, it is how the pit is checked at the handover.

Primary work time: the hours the digger is loading; the only time revenue is made.

Prime, Validation Lab, Apply, Verify: the four stages of the program.

Productivity levers: the two things that move total volume, operating time and rate, and what sits under each (chapter 4).

Queue time: a truck waiting at the face for the one in front.

ROM: run of mine: ore or coal as mined, before processing; the pad it is delivered to.

Safety interaction: a short, two-way talk with an operator at the work about the job in front of them.

Secondary work time: time a machine is working but not loading, such as walking between faces.

Six behaviours: plan, assign, watch, respond, fix and report: what an effective manager does, rated Yes or No each roster.

SMU: service meter unit: the engine-hour meter on a machine. SMU hours are the hours the engine ran; what the plant is charged on.

Superintendents' Lab: half a day in the room, in week 2, on the Superintendent's own model, before the supervisors start.

Swing: a crew's roster period on site, from its first shift to its last.

System (the): the site's electronic record (payloads, cycle times) where it has one. Its figure sits beside the supervisor's.

TARP: trigger action response plan: what to do, set in advance, when a limit is reached.

Technical Services: the site's engineers, planners and surveyors.

Time in field: the supervisor's four blocks, in the pit: at least 90% of the shift.

Time in office: the supervisor's Shift Report and other admin, done in Block 4 and kept short.

Tip head: the edge of the dump where trucks tip.

Touch-base 11:30: 15 minutes with each supervisor, face to face, preferably at the lookout on day shift: Block 2 in numbers and a decision for each area.

Truck allocation: which trucks, and how many, run on each circuit.

Two-week plan: the next two weeks of work, by area and machine, that today's choices affect.

Utilisation: the share of available time the gear is really working, not waiting, walking or parked.

Weekly Commitment Meeting: where next week's plan is signed by the people who will deliver it, once it can be delivered.

Weekly Site Planning Meeting: where next week's plan is made, across departments, with maintenance and blast windows locked.

Weekly Site Review: the site's weekly IN meeting, weeks 1 to 3, chaired by the Project Manager.

What we got away with: a weak signal: something that nearly went wrong and did not, this time.

The evidence behind the model

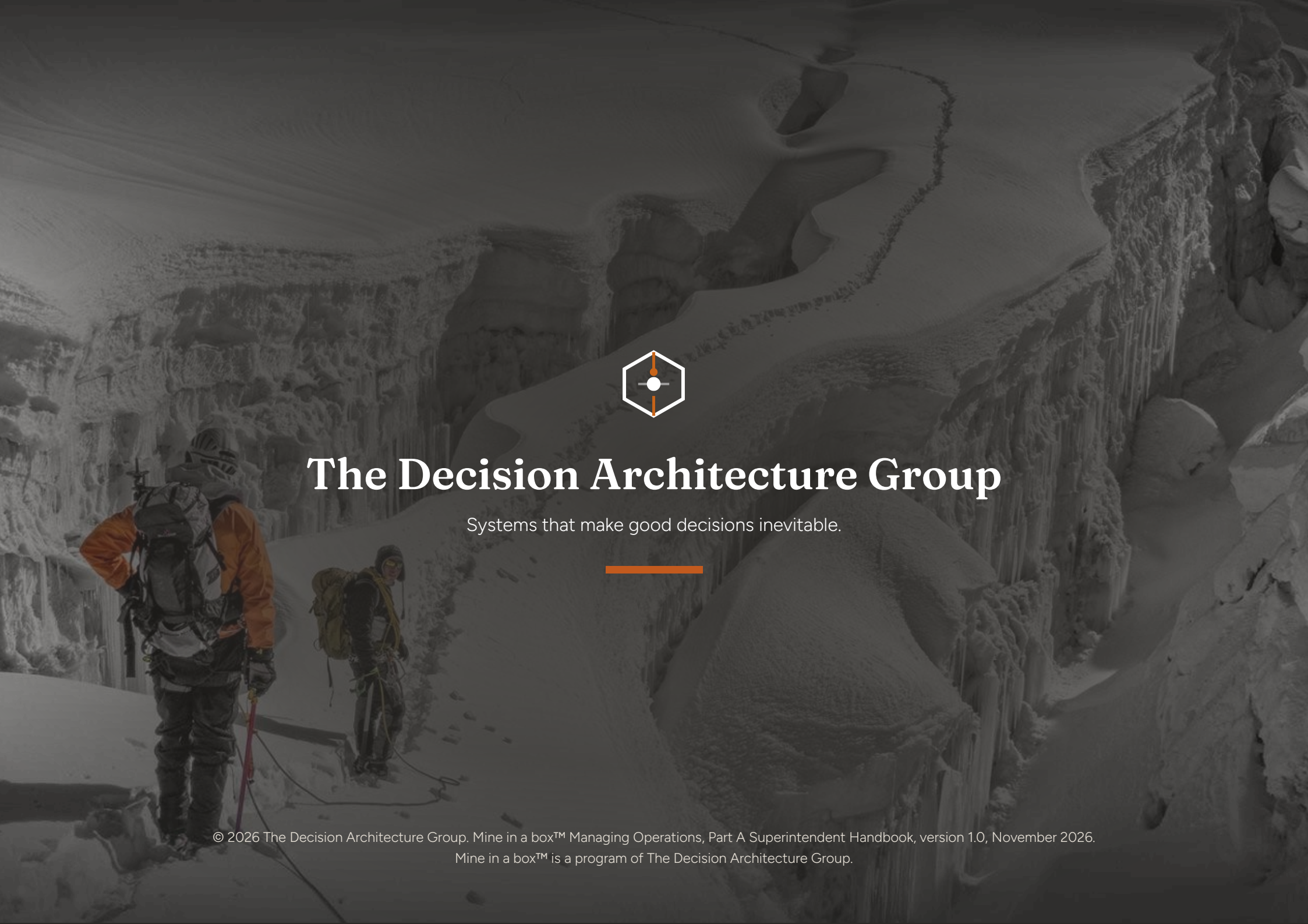
The way the department is run here rests on a few well-tested ideas.

- **Every role has a time span (Jaques 1989; Macdonald, Burke and Stewart 2018).** Requisite organisation and Systems Leadership set each level of work by the time span of its longest task: level II, 3 months to 1 year; level III, 1 to 2 years. That is why the supervisor has the shift and you have the system (chapter 2).
- **A decision made under pressure depends on who makes it (Kahneman, Sibony and Sunstein 2021).** Different people give different answers to the same problem. Placing decisions before the work takes that variation out of the shift.
- **What the supervisor checks, the crew does (Zohar 2002; Zohar and Luria 2003; Zohar and Polachek 2014).** When supervisors got feedback on their safety talks at the work, and were coached on their daily exchanges, safe behaviour rose and injuries fell. That is why you coach from the cards every shift.
- **The best managers watch the work (Komaki 1986, Journal of Applied Psychology; Komaki, Desselles and Bowman 1989).** What set the most effective managers apart was time spent watching the work, so their feedback matched what people had really done. That is why the six behaviours put watching at the centre (chapter 2).
- **Small failures warn of big ones (Weick and Sutcliffe 2015).** High-reliability organisations work to five principles: preoccupation with failure, sensitivity to operations, reluctance to simplify, deference to expertise and commitment to resilience. That is why safety signals are read as a pattern at the Daily Site Review, and why you check the five with each supervisor once a roster.
- **Management makes performance predictable (Kotter 1990).** Management copes with complexity by planning, organising and controlling; its product is order and predictability.

Where it has run. Indicative results from client sites (unpublished): fewer safety incidents, more production and lower staff turnover where the full program is deployed. Results vary by site and by how fully the program is deployed.

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